JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON Friday 23rd JANUARY 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. Let the domain of the function $f(x) = \log_3\log_5\log_7(9x - x^2 - 13)$ be the interval (m, n). Let the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2} = 1$ have eccentricity $\frac{n}{3}$ and the length of the latus rectum $\frac{8m}{3}$. Then $b^2 - a^2$ is equal to :

(1) 5 (2) 11

(3) 9 (4) 7

2. Let $f(x) = \int \frac{(2-x^2).e^x}{\left(\sqrt{1+x}\right)(1-x)^{3/2}}dx$. If $f(0) - 0$, then $f\left(\frac{1}{2}\right)$ is equal to :

(1) $\sqrt{3e}-1$ (2) $\sqrt{2e}+1$

(3) $\sqrt{2e}-1$ (4) $\sqrt{3e}+1$

3. Let $\vec{a}=-\hat{i}+\hat{j}+2\hat{k}$, $\vec{b}=\hat{i}-\hat{j}-3\hat{k}$, $\vec{c}=\vec{a}\times\vec{b}$ and $\vec{d}=\vec{c}\times\vec{a}$. Then $\left(\vec{a}-\vec{b}\right)\cdot\vec{d}$ is equal to :

(1) 4 (2) –4

(3) –2 (4) 2

4. A rectangle is formed by the lines x = 0, y = 0, x = 3 and y = 4. Let the line L be perpendicular to 3x + y + 6 = 0 and divide the area of the rectangle into two equal parts. Then the distance of the point $\left(\frac{1}{2},-5\right)$ from the line L is equal to :

(1) $2\sqrt{5}$ (2) $3\sqrt{10}$

(3) $\sqrt{10}$ (4) $2\sqrt{10}$

5. Let the line y – x = 1 intersect the ellipse $\frac{x^2}{2}+\frac{y^2}{1} = 1$ at the points A and B. Then the angle made by the line segment AB at the center of the ellipse is :

(1) $\pi - \tan^{-1}\left(\frac{1}{4}\right)$ (2) $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$

(3) $\frac{\pi}{2}+2\tan^{-1}\left(\frac{1}{4}\right)$ (4) $\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{4}\right)$

6. Let A={–2, –1, 0,1, 2, 3, 4}. Let R be a relation on A defined by xRy if and only if 2x+ y ≤ 2. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then l + m + n is equal to :

(1) 32 (2) 34

(3) 33 (4) 35

7. Let y = y(x) be the solution of the differential equation

$x^4dy + (4x^3y + 2\sin x)dx = 0$, $x > 0$, $y\left(\frac{\pi}{2}\right) = 0$.

Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

(1) 81 (2) 92

(3) 64 (4) 72

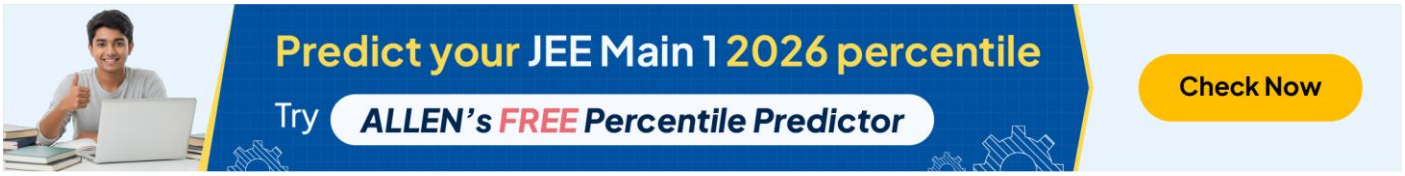
8. If α and β (α < β) are the roots of the equation

$\left(-2+\sqrt{3}\right)\left(\left|\sqrt{x}-3\right|\right)+\left(x-6\sqrt{x}\right)+\left(9-2\sqrt{3}\right) = 0$,

$x \geq 0$, then $\sqrt{\frac{\beta}{\alpha}}+\sqrt{\alpha\beta}$ is equal to :

(1) 8 (2) 9

(3) 10 (4) 11



9. Let $f(x) = \begin{cases} \frac{ax^2+2ax+3}{4x^2+4x-3}, & x \neq -\frac{3}{2}, \frac{1}{2} \\ b, & x = -\frac{3}{2}, \frac{1}{2} \end{cases}$ be continuous at $x = -\frac{3}{2}$. If $fof(x) = \frac{7}{5}$, then x is equal to :

(1) 2 (2) 1

(3) 0 (4) 1.4

10. The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1+\sqrt[3]{\tan 2x}}$ is :

(1) $\frac{\pi}{12}$ (2) $\frac{\pi}{18}$

(3) $\frac{\pi}{6}$ (4) $\frac{\pi}{3}$

11. Among the statements :

I : If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$, and

II : If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = px + q$, then $p^2 = 196q^2$,

(1) both are false (2) only II is true

(3) both are true (4) only I is true

12. The value of $\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \ldots + \frac{{}^{100}C_{100}}{101}$ is :

(1) $\frac{2^{101}}{100}$ (2) $\frac{2^{100}}{100}$

(3) $\frac{2^{101}}{101}$ (4) $\frac{2^{100}}{101}$

13. Let the mean and variance of 8 numbers –10, –7, –1, x, y, 9, 2, 16 be $\frac{7}{2}$ and $\frac{293}{4}$, respectively. Then the mean of 4 numbers x, y, x + y + 1, |x – y| is:

(1) 11 (2) 9

(3) 10 (4) 12

14. The sum of all possible values of $n \in N$, so that the coefficients of x, x^2 and x^3 in the expansion of $(1 + x^2)^2 (1 + x)^n$, are in arithmetic progression is :

(1) 3 (2) 7

(3) 12 (4) 9

15. The vertices B and C of a triangle ABC lie on the line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$. The coordinates of A and B are (1, 6, 3) and (4, 9, α) respectively and C is at a distance of 10 units from B. The area (in sq. units) of ΔABC is:

(1) $5\sqrt{13}$ (2) $15\sqrt{13}$

(3) $20\sqrt{13}$ (4) $10\sqrt{13}$

16. Number of solutions of $\sqrt{3}\cos 2\theta + 8\cos\theta + 3\sqrt{3} = 0, \theta \in [-3\pi, 2\pi]$ is:

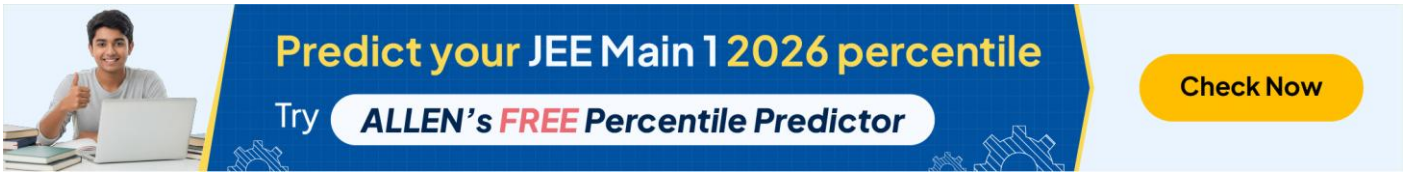
(1) 0 (2) 5

(3) 3 (4) 4

17. Let $S = \{z : 3 \leq |2z - 3(1 + i)| \leq 7\}$ be a set of complex numbers. Then $\min_{z \in S} \left|\left(z + \frac{1}{2}(5 + 3i)\right)\right|$ is equal to :

(1) $\frac{1}{2}$ (2) $\frac{3}{2}$

(3) 2 (4) $\frac{5}{2}$



18. Let α and β respectively be the maximum and the minimum values of the function

$$f(\theta) = 4\left(\sin^4\left(\frac{7\pi}{2}-\theta\right)+\sin^4(11\pi+\theta)\right)$$

$$-2\left(\sin^6\left(\frac{3\pi}{2}-\theta\right)+\sin^6(9\pi-\theta)\right), \theta \in \mathbf{R}.$$

Then $\alpha + 2\beta$ is equal to :

(1) 4 (2) 5

(3) 3 (4) 6

19. A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. Then mason A alone will complete the construction work in:

(1) 24 days (2) 42 days

(3) 30 days (4) 36 days

20. Let the direction cosines of two lines satisfy the equations : $4\ell + m - n = 0$ and $2mn + 10n\ell + 3\ell m = 0$. Then the cosine of the acute angle between these lines is :

(1) $\frac{10}{\sqrt{38}}$ (2) $\frac{20}{3\sqrt{38}}$

(3) $\frac{10}{7\sqrt{38}}$ (4) $\frac{10}{3\sqrt{38}}$

SECTION-B

21. Let $|A| = 6$, where A is a 3×3 matrix. If $|\text{adj}(3\text{adj}(A^2 \cdot \text{adj}(2A)))| = 2^m.3^n$, m, n $\in$ **N**, then m + n is equal to _____.

22. Let the area of the region bounded by the curve $y = \max\{\sin x, \cos x\}$, lines $x = 0$, $x = \frac{3\pi}{2}$, and the x-axis be A. Then, $A + A^2$ is equal to

23. Let f be a twice differentiable non-negative function such that

$(f(x))^2 = 25 + \int_0^x \left((f(t))^2 + (f'(t))^2\right)dt$. Then the mean of $f(\log_e(1))$, $f(\log_e(2))$, ..., $f(\log_e(625))$ is equal to _____ .

24. From the first 100 natural numbers, two numbers first a and then b are selected randomly without replacement. If the probability that $a - b \geq 10$ is $\frac{m}{n}$, gcd (m, n) = 1, then m + n is equal to _____.

25. The number of 4-letter words, with or without meaning, which can be formed using the letters PQRPQRSTUVP, is _____ .



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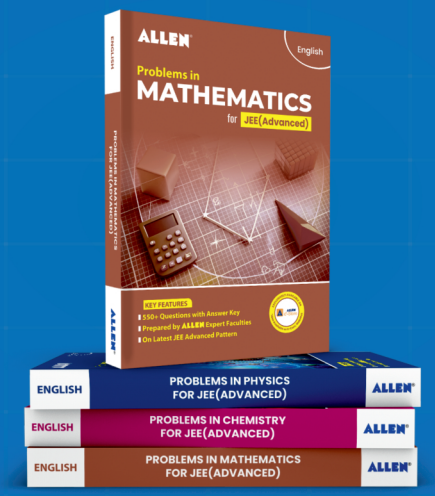


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